The Decoupled Threshold VAR: Mean and Variance Regimes in State-Dependent Fiscal Multipliers

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Introduction

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Does the volatility of shocks matter for regime dependence in macro?

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Macro dynamics and macro volatility both change over time

- Great Moderation: large fall in the volatility of macro shocks
- U.S. monetary policy regimes: best fit when only variances switch
- Uncertainty, oil and fiscal volatility shocks have real effects

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How time variation is modelled

1. TVP-VARs with stochastic volatility, Markov switching, ST-VAR
   - TVP and MS: changes are latent, their timing exogenous; heavy to estimate
   - ST-VAR: observable state, calibrated transition (AG2012)
2. TVARs: observable state, estimated threshold
   - parsimonious: precise estimates, regimes easy to interpret
   - ! but their estimation theory uses the **conditional mean only**

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This paper: volatility regimes in a threshold VAR, uncovering novel regime dependence in fiscal multipliers

Contributions

1. A threshold VAR in which volatility has regimes of its own
2. Estimation theory for the thresholds
3. New regime dependence in U.S. fiscal multipliers

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1. A threshold VAR in which volatility has regimes of its own

$$y_t = \underbrace{\sum_{i=1}^{k_\mu} \Phi_i x_t \, \mathbb{I}\left(q^{\mu}_{t-1} \in \mathcal{R}^{\mu}_i\right)}_{\text{propagation: mean regimes}} + \underbrace{\sum_{\ell=1}^{k_\sigma} \Sigma_\ell^{1/2} \, \mathbb{I}\left(q^{\sigma}_{t-1} \in \mathcal{R}^{\sigma}_\ell\right) \varepsilon_t}_{\text{shock environment: variance regimes}}$$

   - **single** structure pools two shock environments: wrong **impact IRFs**
   - a **common** structure compares shock environments, not states
   - here each switch is located separately: own state and threshold for mean and variance
   - the common structure: a special case, which the data can select

2. Estimation theory for the thresholds

3. New regime dependence in U.S. fiscal multipliers

Contributions

1. A threshold VAR in which volatility has regimes of its own
2. Estimation theory for the thresholds
   - the standard (least-squares) estimator cannot see a regime change in the variance: inconsistent
   - likelihood-based estimator: consistent; thresholds converge at rate T, with nonstandard limits
   - standard $\sqrt{T}$ inference on everything else $\Rightarrow$ estimating the thresholds is first-order asymptotically costless
3. New regime dependence in U.S. fiscal multipliers

Contributions

1. A threshold VAR in which volatility has regimes of its own

2. Estimation theory for the thresholds

3. New regime dependence in U.S. fiscal multipliers

   Q: do spending multipliers depend on the state of the economy?

   A: The literature disagrees: larger in recessions (AG2012), no different (RZ2018), larger in expansions (Alloza2022)

   ! All three **calibrate** the regimes

   $\Rightarrow$ I estimate them:

   - wars and military build-ups form a volatility regime of their own
   - once isolated, state dependence even at RZ2018's own threshold
   - the direction depends on the sample: larger in high unemployment with pre-WWII data, in low unemployment after WWII

Related Literature

Threshold models

- Tong and Lim (1980), Chan (1993), Tsay (1998), Hansen (2000), Samia and Chan (2011), Gargiulo, Matthes and Petrova (2026); STVAR: Teräsvirta (1994), Auerbach and Gorodnichenko (2012)

⇒ *my paper:* multivariate, separate mean and variance regimes, both estimated

Volatility and latent regimes

- Hamilton (1989), Cogley and Sargent (2005), Primiceri (2005), Sims and Zha (2006), Bloom (2009), Fernández-Villaverde et al. (2015)

⇒ *my paper:* volatility regimes on observable states, estimated thresholds

State-dependent fiscal multipliers

- Auerbach and Gorodnichenko (2012), Owyang, Ramey and Zubairy (2013), Fazzari et al. (2015), Ramey and Zubairy (2018), Alloza (2022), Ghassibe and Zanetti (2022), Barnichon, Debortoli and Matthes (2022)

⇒ *my paper:* the regime structure is estimated, not calibrated

Outline

1. TVAR setup and illustration
2. The decoupled threshold VAR: model, estimator, theory
3. Application: data-driven regimes in Ramey and Zubairy (2018)

Setup and Illustration

How Is a Threshold VAR Estimated?

Standard threshold VAR, two regimes: the dynamics switch when an observable state q_{t-1} crosses the threshold γ

$$y_t = \Phi_1 x_t \, \mathbb{I}(q_{t-1} \leq \gamma) + \Phi_2 x_t \, \mathbb{I}(q_{t-1} > \gamma) + \Sigma^{1/2} \varepsilon_t, \qquad x_t = (1, y'_{t-1}, \ldots, y'_{t-p})'$$

Estimation in two steps

1. **Threshold**: for each candidate γ on a grid, split the sample, run OLS in each regime and compute the sum of squared residuals

$$\hat{\gamma} = \arg\min_{\gamma \in \Gamma} SSR(\gamma)$$

2. **Regime parameters**: OLS within each regime, given $\hat{\gamma}$

- $\hat{\gamma}$ converges at rate T: estimation error does not affect inference on Φ_i

  $\Rightarrow$ IRFs by regime, as in a linear VAR

! $SSR(\gamma)$ depends on the conditional mean only: a regime change in the variance is invisible, even if Σ is allowed to differ in step 2

The Importance of Using the Likelihood

Same grid search, different objective: **Gaussian likelihood** with regime-specific covariances $\Rightarrow$ uses breaks in means and variances

RMSE of the threshold estimate (Monte Carlo)

	μ switches, σ constant	
	$T = 200$	$T = 1000$
SSR	0.025	0.004
Likelihood	0.025	0.004

- Only the mean switches: nothing is lost with the likelihood

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SSR	0.025	0.004	**0.081**	0.010
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- Both switch: the likelihood halves the error at $T = 200$

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RMSE of the threshold estimate (Monte Carlo)

	μ switches, σ constant		μ and σ switch		σ switches, μ constant	
	$T = 200$	$T = 1000$	$T = 200$	$T = 1000$	$T = 200$	$T = 1000$
SSR	0.025	0.004	0.081	0.010	**2.004**	**3.204**
Likelihood	0.025	0.004	0.039	0.006	**0.080**	**0.013**

- Only the mean switches: nothing is lost with the likelihood
- Both switch: the likelihood halves the error at $T = 200$
- Only the variance switches (the Sims–Zha case): SSR is inconsistent, the likelihood still converges fast

$\Rightarrow$ Volatility regimes enter threshold model **through the likelihood**

Mean and Variance Regimes in a Threshold VAR

The Model

Threshold VAR(p) for $y_t \in \mathbb{R}^n$, with k_μ mean regimes and a **constant variance**

$$y_t = \underbrace{\sum_{i=1}^{k_\mu} \left(\Phi_{i,0} + \sum_{j=1}^{p} \Phi_{i,j}\, y_{t-j} \right) \mathbb{I}\left(q^{\mu}_{t-d_\mu} \in \mathcal{R}^{\mu}_{i}\right)}_{\text{propagation: mean regimes}} + \underbrace{\Sigma^{1/2}\, \varepsilon_t}_{\text{constant variance}}$$

- $\varepsilon_t \sim NID(0, I_n)$ as estimation criterion, not an assumption; observable state, known at $t-1$: q^{μ}_{t}
- Thresholds define the regimes $\mathcal{R}^{\mu}_{i} = [\gamma^{\mu}_{i-1}, \gamma^{\mu}_{i})$

The Model

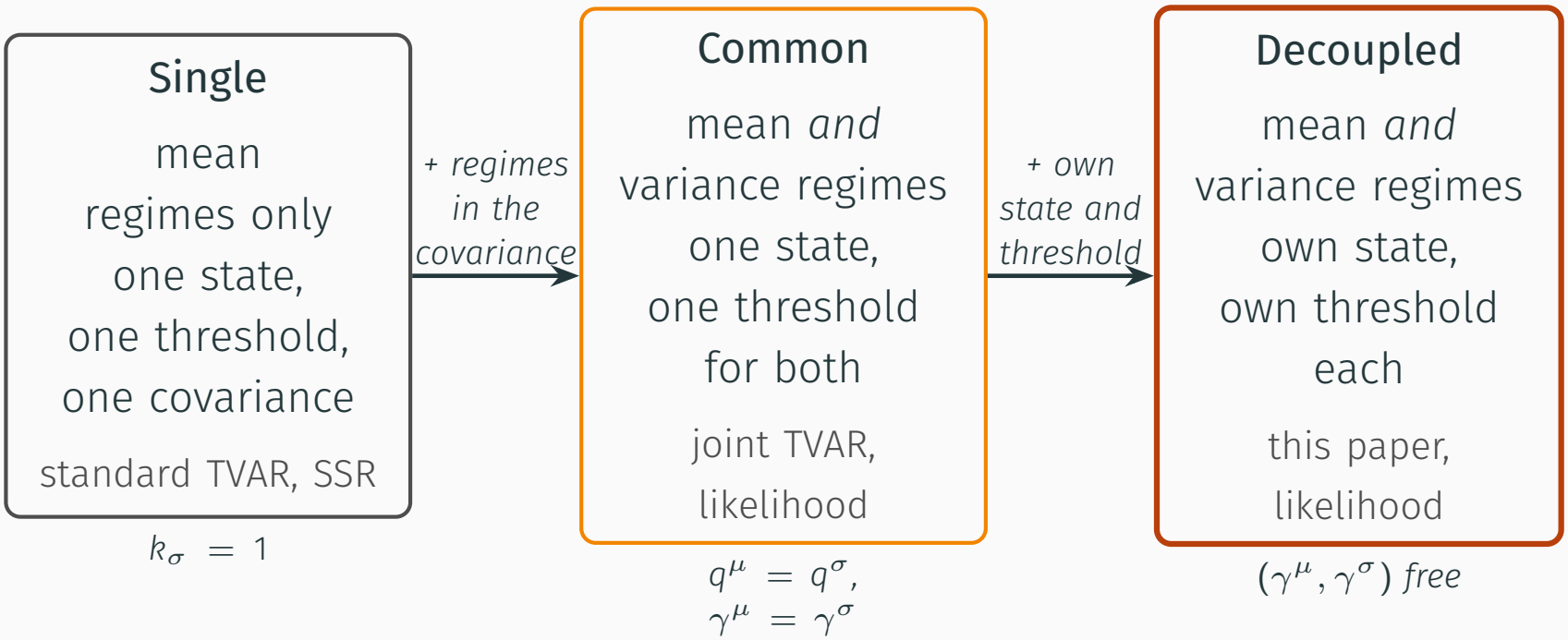
Decoupled threshold VAR(p) for $y_t \in \mathbb{R}^n$, with k_μ mean regimes and k_σ **variance regimes**

$$y_t = \underbrace{\sum_{i=1}^{k_\mu} \left(\Phi_{i,0} + \sum_{j=1}^{p} \Phi_{i,j}\, y_{t-j} \right) \mathbb{I}\left(q^{\mu}_{t-d_\mu} \in \mathcal{R}^{\mu}_i\right)}_{\text{propagation: mean regimes}} + \underbrace{\sum_{\ell=1}^{k_\sigma} \Sigma_\ell^{1/2}\, \mathbb{I}\left(q^{\sigma}_{t-d_\sigma} \in \mathcal{R}^{\sigma}_\ell\right) \varepsilon_t}_{\text{shock environment: variance regimes}}$$

- $\varepsilon_t \sim NID(0, I_n)$ as estimation criterion, not an assumption; observable states, known at $t-1$: q^{μ}_t and q^{σ}_t
- Thresholds define the regimes $\mathcal{R}^{\mu}_i = [\gamma^{\mu}_{i-1}, \gamma^{\mu}_i)$, $\mathcal{R}^{\sigma}_\ell = [\gamma^{\sigma}_{\ell-1}, \gamma^{\sigma}_\ell)$; each observation is in a cell (i, ℓ)

Special cases: $k_\sigma = 1$, the standard TVAR; $q^\mu = q^\sigma$ and $\gamma^\mu = \gamma^\sigma$, the joint TVAR

Three Regime Structures



- each structure nests next one; data can test the first restriction and select the second
- the threshold criterion from SSR to the likelihood: incorporating covariance info

Likelihood-based Estimator

For each candidate pair $(\gamma^{\mu}, \gamma^{\sigma})$, the Gaussian likelihood gives the regime parameters in closed form

- $\hat{\Phi}_i$: ML given the covariances **is GLS** within mean regime i, each observation weighted by the precision $\hat{\Sigma}_{\ell}^{-1}$ of its cell
- $\hat{\Sigma}_{\ell}$: ML given the coefficients, residual covariance pooled within variance regime ℓ
- one-step feasible version (OLS $\to \hat{\Sigma}_{\ell}^{(0)} \to$ GLS $\to$ one update) is first-order equivalent to full profile ML

Threshold estimator: maximize the profile likelihood over a grid

$$(\hat{\gamma}^{\mu}, \hat{\gamma}^{\sigma}) = \arg \max_{(\gamma^{\mu}, \gamma^{\sigma}) \in \Gamma^{\mu} \times \Gamma^{\sigma}} -\frac{1}{2} \sum_{\ell=1}^{k_{\sigma}} \frac{t_{\ell}}{T} \ln \det \hat{\Sigma}_{\ell}(\gamma^{\mu}, \gamma^{\sigma})$$

- Two steps, as in any threshold model

Algorithm

! Thresholds identified from breaks in conditional mean **and** covariance

Assumptions

1. Stability (A1–A4): geometric ergodicity, finite fourth moments — Theorem A1
   - ε_t i.i.d., positive density, $E\|\varepsilon_t\|^4 < \infty$; $\Sigma_\ell \succ 0$ — A1–A4
   - common Lyapunov matrix for the **mean** regimes only: $A_i' P A_i \preceq \lambda_i^2 P$ (sufficient, not necessary) — A4 at the estimates
2. Identification (B1–B4, B9) — B1–B4, B9
   - Θ compact, θ_0 interior; $f_\mu(\gamma_i^{\mu,0}) > 0, f_\sigma(\gamma_\ell^{\sigma,0}) > 0$
   - $\Phi_i \neq \Phi_{i+1}$ at the threshold; $\Sigma_\ell \neq \Sigma_{\ell+1}$
   - every cell (i, ℓ) and every candidate regime has positive probability
3. Regularity local to the thresholds (B5–B8) — B5–B8
   - weak dependence, bounded conditional moments, continuity near $\gamma^{\mu,0}$, $\gamma^{\sigma,0}$
   - ! B8, for the joint limit law only: no excess mass near *both* cutoffs at once (automatic for distinct coordinates; shared state with equal cutoffs = the common structure)

Results

Theorem 1 (consistency). Under 1–2:

$$\hat{\theta}_T \to_{a.s.} \theta_0 \qquad \text{as} \quad T \to \infty, \qquad \theta = \left(\gamma^{\mu}, \gamma^{\sigma}, \{\Phi_i\}, \{\Sigma_\ell\}\right)$$

- Extremum-estimator argument on the feasible profile criterion: ULLN, unique population maximizer, argmax theorem
- Gaussianity is the criterion

Theorem 1

Results

Theorem 2 (rates and limit laws). Under 1–3:

(i) **Rates**: $T(\hat{\gamma}^{\mu} - \gamma^{\mu,0}) = O_p(1)$, $T(\hat{\gamma}^{\sigma} - \gamma^{\sigma,0}) = O_p(1)$; regime parameters at $\sqrt{T}$

(ii) **Thresholds**: asymptotically independent compound-Poisson argmax limits

(iii) **Regime parameters**: Gaussian, as if the thresholds were known; $\hat{\Phi}_i$ at the GLS bound

$\Rightarrow$ **Estimating the thresholds is first-order asymptotically costless**: under any identification, IRFs and multipliers are smooth functions of $(\hat{\Phi}_i, \hat{\Sigma}_\ell)$ and inherit $\sqrt{T}$ inference given $(\hat{\gamma}^{\mu}, \hat{\gamma}^{\sigma})$

- Threshold limits not pivotal (no bootstrap for the thresholds themselves); fully parametric plug-in CIs for $\gamma^{\sigma,0}$ Part (i) Parts (ii)–(iii) Inference

Special Cases and Nesting

What the **estimator** does on each rung:

- **Single, variance-only, linear**: the absent threshold is not identified $\Rightarrow$ restriction tested (sup-LR, bootstrap p-values, Hansen 1996)
- **Common (joint TVAR)**: both thresholds identified $\Rightarrow$ restriction selected

Joint model

Proposition 1 (nesting). Unrestricted estimator, shared state; common structure true:

(i) $\hat{\gamma}^{\mu}$, $\hat{\gamma}^{\sigma}$ both reach the common threshold at rate T; (ii) regime parameters as in the joint estimator

(iii) the gap $T\big(\hat{L}_T^F - \hat{L}_T^J\big)$ stays bounded; it grows at rate T if the thresholds differ

$\Rightarrow$ select the joint model if $T\big(\hat{L}_T^F - \hat{L}_T^J\big) < c_T$; consistent for any $c_T \to \infty$, $c_T/T \to 0$ ($c_T \propto \ln T$)

Monte Carlo: joint model selected in **99–100%** of samples when true, **0%** when decoupled

Table

Monte Carlo: What a Common Structure Misses

True model: mean regimes driven by one state, variance regimes by another

RMSE of thresholds and regime parameters

	$T = 200$				$T = 1000$			
Estimator	$\hat{\gamma}^{\mu}$	$\hat{\gamma}^{\sigma}$	$\hat{\Phi}$	$\hat{\Sigma}$	$\hat{\gamma}^{\mu}$	$\hat{\gamma}^{\sigma}$	$\hat{\Phi}$	$\hat{\Sigma}$
Decoupled	0.040	0.103	0.38	0.83	0.008	0.018	0.16	0.36
Common on q^{μ}	0.092	–	0.46	2.21	0.009	–	0.19	2.13
Common on q^{σ}	–	0.138	1.62	1.17	–	0.022	1.55	0.88
Single (SSR) on q^{μ}	0.051	–	0.45	2.17	0.008	–	0.19	2.13

- **Decoupled**: smallest error in every column, all errors shrink with T
- **Common or single structure**: the dimension not modelled is never learned, its error does not shrink with T
- **SSR on the mean state** is what a practitioner would run: fine for γ^{μ}, silent on the variance regimes, pooled $\hat{\Sigma}$ far from both true ones

Limiting distributions

Application: Data-driven Regimes in Ramey and Zubairy (2018)

The Application: State-Dependent Fiscal Multipliers

Q: Do the effects of fiscal shocks depend on the state of the economy?

A: The literature disagrees. Government spending multipliers are

1. larger in recessions — AG2012
2. no different across states — RZ2018
3. larger in expansions — Alloza2022

! In all three the regimes are calibrated: the researcher picks the state variable and the cut-off

I take an agnostic approach: estimate the regimes

- thresholds estimated from the data, not calibrated
- fiscal shocks come with wars and large build-ups: a natural place for volatility regimes

$\Rightarrow$ Is the disagreement due to how the **regimes are specified**?

Setup

RZ2018: state-dependent local projections (LP-IV) for the government spending multiplier

- Output y_t and spending g_t over potential; instrument: military news shock z_t Weak IV
- State: lagged unemployment, threshold calibrated at 6.5%
- Samples: full (1890Q1–2015Q4), excluding WWII, post-WWII
- Result: **no state dependence**, 0.60 (high-u) vs 0.59 (low-u), $p = 0.95$

What I do: same data, same state variable, but the regimes are estimated

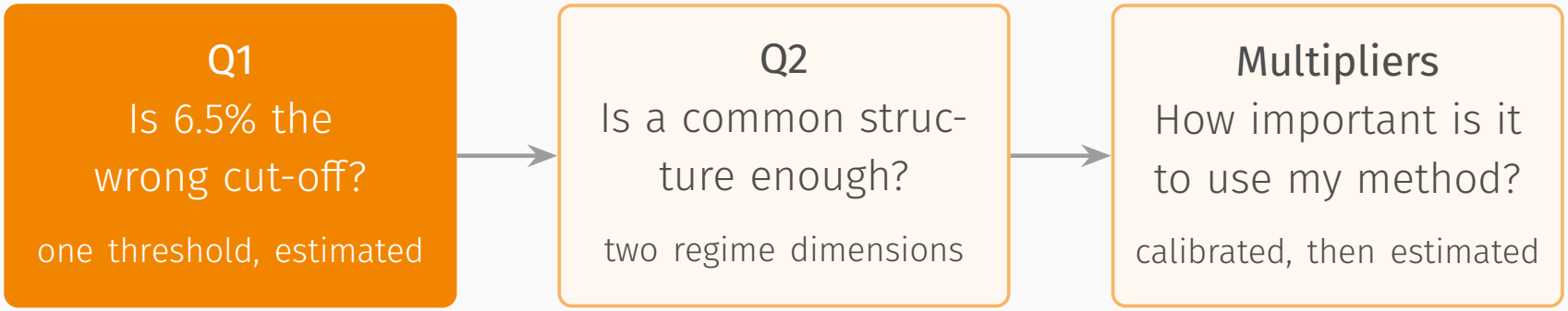
- TVAR in $[z_t, y_t, g_t]'$, instrument ordered first: same IRFs as the LP-IV
  - Companion project: equivalence conditions for threshold estimation in LP-IV

Q1: Is 6.5% the wrong cut-off?

Q2: Is a common regime structure enough?

RZ2018's interest-rate (ZLB) exercise gives the same picture Interest-rate exercise

The Application: Roadmap



- Start where the literature is: **one** threshold on unemployment, RZ2018's calibrated 6.5% against the estimate

Q1: Is 6.5% the Wrong Cut-off?

One threshold on lagged unemployment, **estimated** with the likelihood

	Estimated (likelihood)		Calibrated (RZ2018)	
Sample	$\hat{\gamma}$ (%)	low-u quarters	γ (%)	low-u quarters
Full sample	3.56	57 (11%)	6.50	319 (64%)
Excl. WWII	3.64†	49 (10%)	6.50	300 (63%)
Post-WWII	5.61	139 (51%)	6.50	189 (70%)

†Smallest regime allowed by the trimming.

All estimators

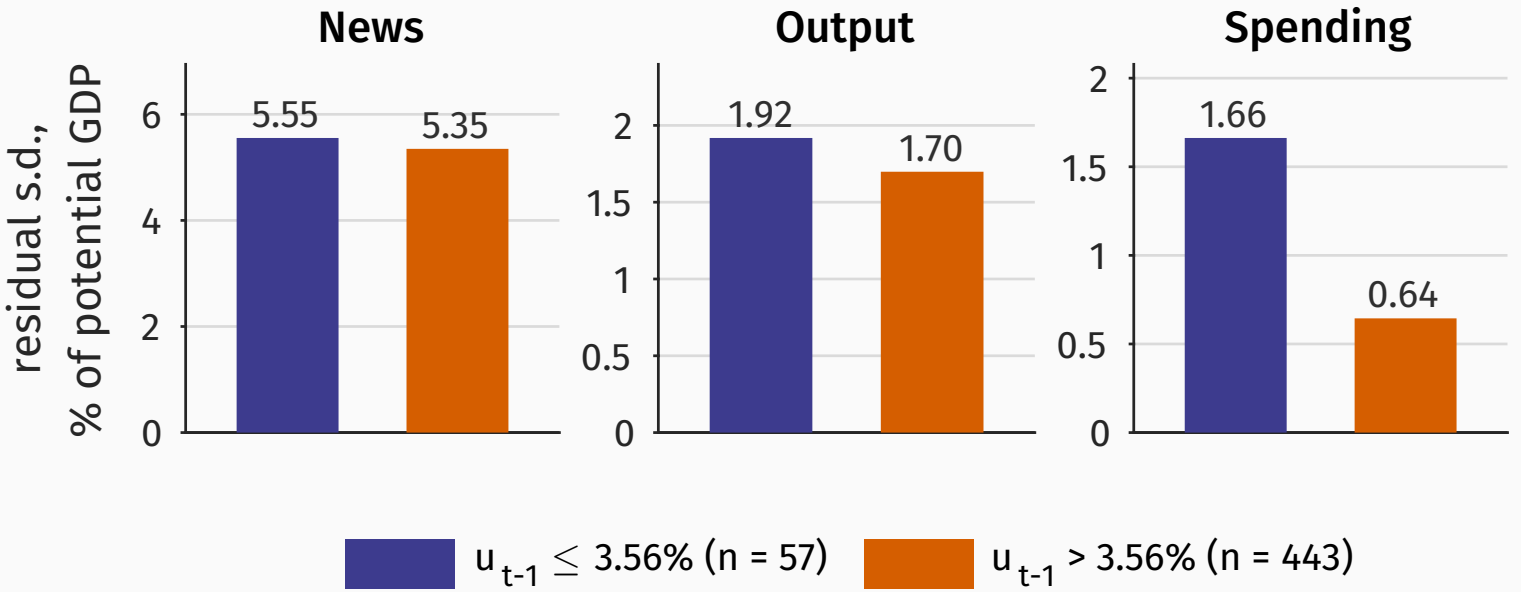
- Full sample: 3.56%, a low-unemployment regime of 57 quarters: wartime full employment and booms

⇒ Not recession/expansion split: a common regime structure **conflates the economic state with the nature of the shocks**

- Post-WWII, no war in the sample: 5.61%, balanced split, closer to RZ2018

⇒ Q1: not a different line, but a second regime dimension

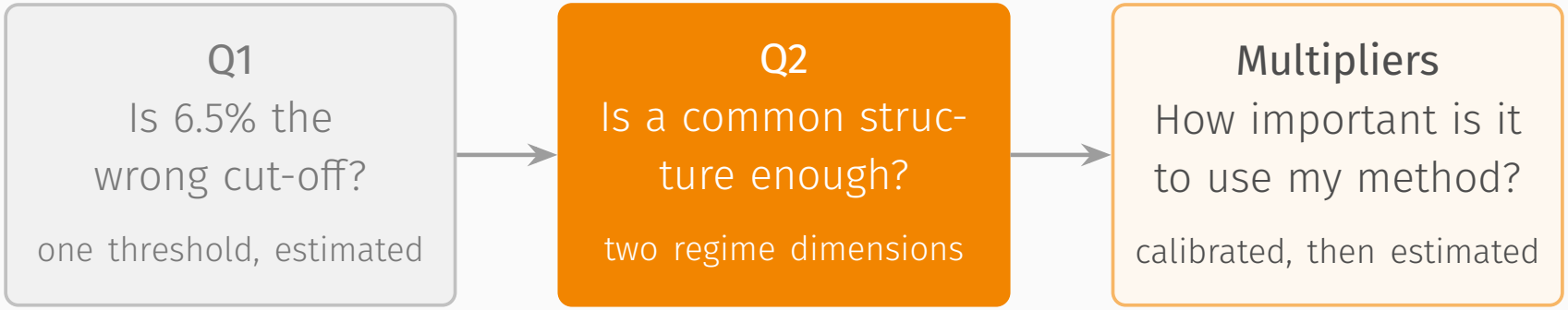
What Does the Likelihood See?



- Volatility of the innovations in the two regimes split at 3.56%
- The break is in **government spending** equation: ratio of 2.6
- Output and news barely change

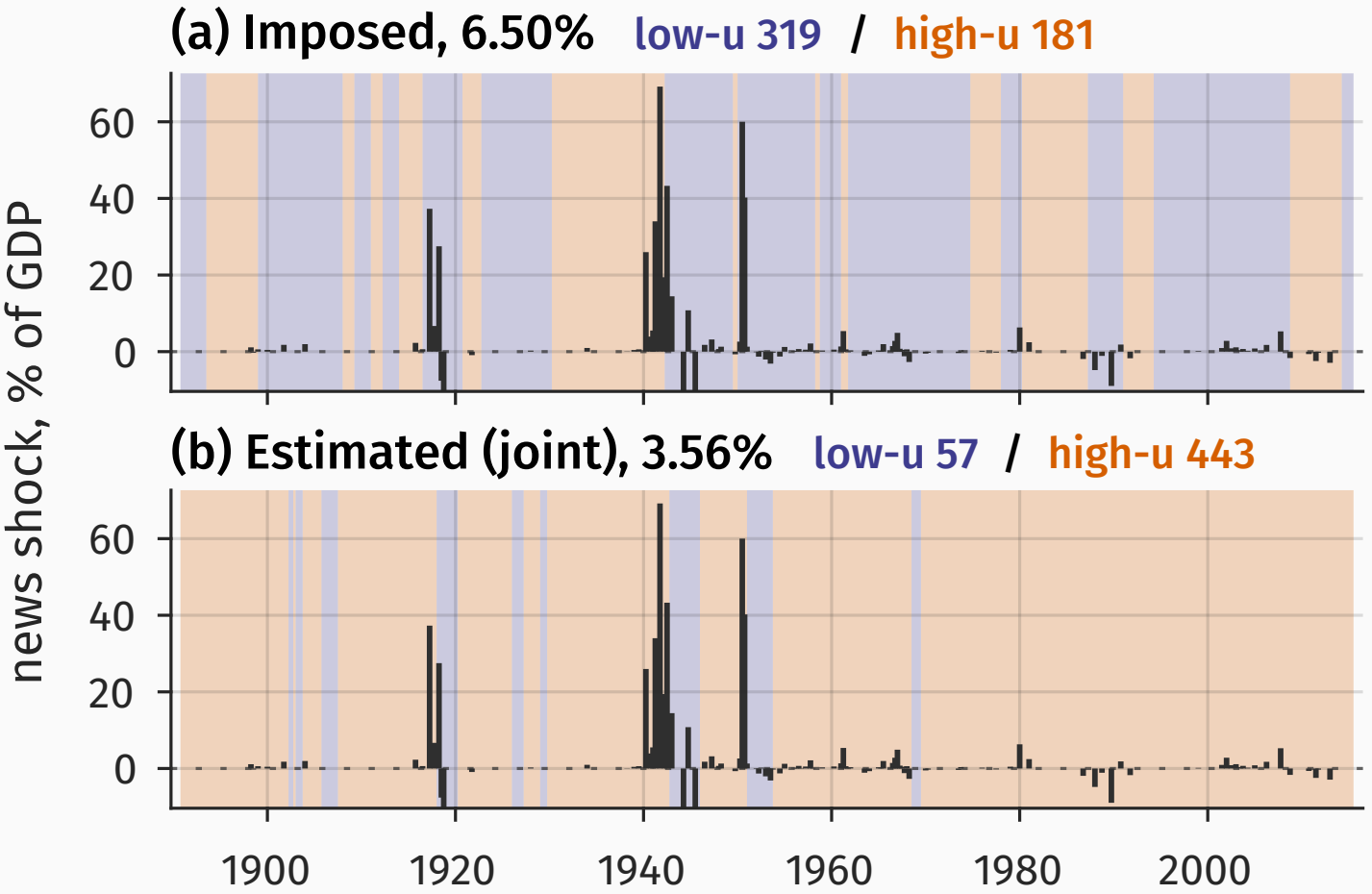
⇒ Threshold sits where the volatility of fiscal innovations changes (wartime spending surges), not where the shock propagation changes

The Application: Roadmap



- ✓ Q1: the estimate, 3.56%, is not a different recession line; it sits where the shocks change
- ▷ The regime needs a **second dimension**: a state and a threshold of its own for the variance

Q2: Is a Common Regime Structure Enough?



- Low-unemployment regime at 3.56%: 11% of quarters, 22% of the instrument's variation; instrument and regime nearly collinear

⇒ Volatility regime forced onto the unemployment split: give it **its own state and threshold**

Two Regime Dimensions

1. **Mean regimes**: lagged unemployment $u_{t-1} \Rightarrow$ low- and high-unemployment regime
2. **Variance regimes**: size of the lagged news shock $|z_{t-1}| \Rightarrow$ **normal-shock** and **fiscal-event** regime

Why not spending itself? It is the response to the news, and lags it

Details

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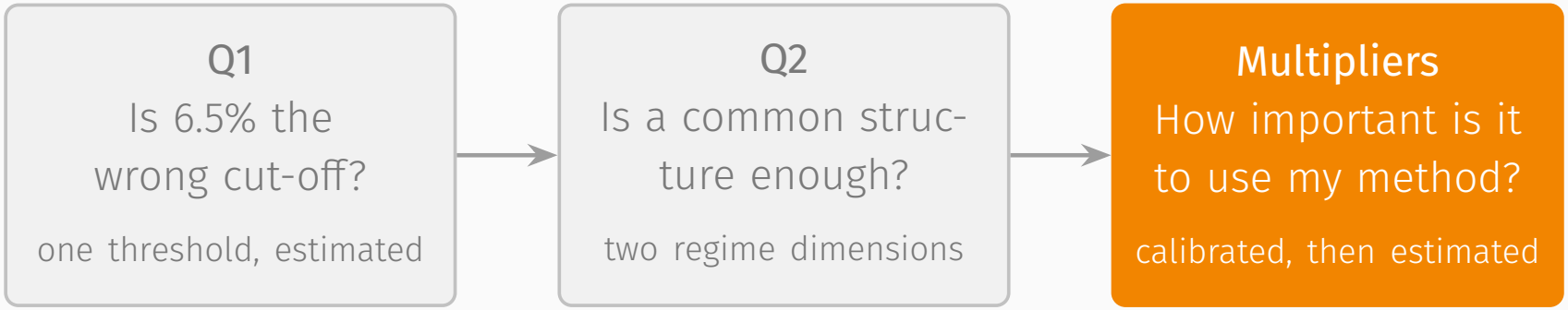
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Details

	Estimated thresholds		Normal-shock obs.		Fiscal-event	$\lvert z \rvert$ share of
Sample	$\hat{\gamma}^{\mu}$ (%)	$\hat{\gamma}^{\sigma}$	low-u	high-u	obs.	fiscal events
Full sample	**8.42**	0.0243	382	84	**33**	37%
Post-WWII	**4.23**	0.0211	47	208	17	53%

- $\hat{\gamma}^{\sigma} \approx 0.024$ in every sample. **Fiscal events**: both World Wars, Korea, Berlin 1961, Vietnam, Reagan build-up, 1988/1990, 2002, 2008–13
- With them isolated, the mean threshold moves to where propagation changes: 8.42% (full sample), 4.23% (post-WWII)

The Application: Roadmap



- ✓ Q1–Q2: two regime dimensions; $|z_{t-1}|$ isolates the **fiscal-event quarters**, the mean threshold moves
- ▷ Multipliers: first at RZ2018's 6.5% with the variance regimes added, then at the **estimated** thresholds

How Variance Regimes Enter the Multiplier

Cumulative multiplier: cumulated responses of output over spending, to the news shock

$$M(h) = \frac{\sum_{s=0}^{h} \phi_{Y,s}}{\sum_{s=0}^{h} \phi_{G,s}}, \qquad \phi_s = \underbrace{\Psi_s(\Phi_i)}_{\text{propagation: mean regime } i} \times \underbrace{b_\ell}_{\text{impact vector: } \Sigma_\ell}$$

- The variance regime affects point estimate **through impact vector only**
  - $M(h)$ is a ratio: the size of the shock cancels, relative impact responses matter
  - at $h = 0$ the multiplier is the same in both unemployment regimes

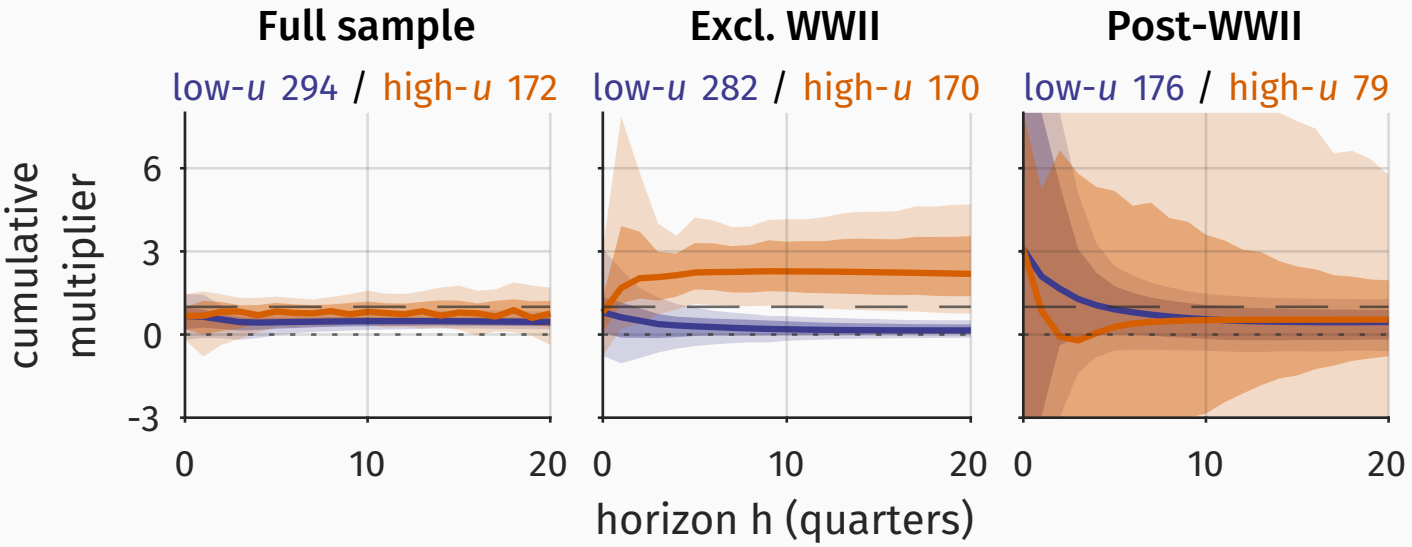
⇒ State dependence = low- vs high-unemployment regime, **within the normal-shock regime**

- test: $\Delta(h) = M_{\text{high-}u}(h) - M_{\text{low-}u}(h)$, 90% bootstrap interval; fiscal-event cells are too small for inference (4 to 28 quarters)

! Without variance regimes, each unemployment regime's impact vector and coefficients average over normal-shock and fiscal-event quarters

How Important Is It to Use My Method?

Keep RZ2018's 6.5% and change **one thing**: add the estimated variance regimes (normal-shock regime, 68% dark and 90% light bands)



- RZ2018at the same threshold, without variance regimes: 0.60 vs 0.59

  VAR vs LP-IV

- **Excl. WWII**: high-unemployment 2.07–2.28 vs low-unemployment 0.15–0.38
- Full sample: same direction, smaller gap. Post-WWII: nothing identified

  Full vs excl. WWII vs estimated

How Important Is It to Use My Method? (2)

Cumulative multipliers at the calibrated 6.5% threshold, normal-shock regime

	$h = 3$	$h = 7$	$h = 11$	$h = 19$
Excl. WWII				
low unemployment	0.38	0.24	0.18	0.15
high unemployment	2.07	2.26	2.28	2.20
Δ (high − low)	**1.69**	**2.01**	**2.10**	**2.05**
90% interval for Δ	*[0.77, 3.37]*	*[0.86, 3.56]*	*[0.75, 3.95]*	*[0.59, 4.53]*
Full sample				
low unemployment	0.45	0.46	0.48	0.46
high unemployment	0.83	0.77	0.78	0.60
Δ (high − low)	0.38	0.30	0.29	0.14
90% interval for Δ	*[-0.29, 0.76]*	*[-0.16, 0.81]*	*[-0.31, 0.99]*	*[-0.63, 1.31]*
Post-WWII: Δ (high − low)	−1.50	−0.27	0.01	0.09
90% interval for Δ	*[-24.5, 20.9]*	*[-14.4, 17.8]*	*[-9.1, 13.1]*	*[-4.3, 5.6]*

- Excl. WWII: the difference is **significant at every horizon**

Weak IV

⇒ Historical samples: the RZ2018 null comes from **ignoring the variance regimes**, not from where the threshold is

Multipliers at the Estimated Thresholds

Full sample

high-u: 1.24 $\rightarrow$ 0.98

low-u: 0.44 $\rightarrow$ 0.45

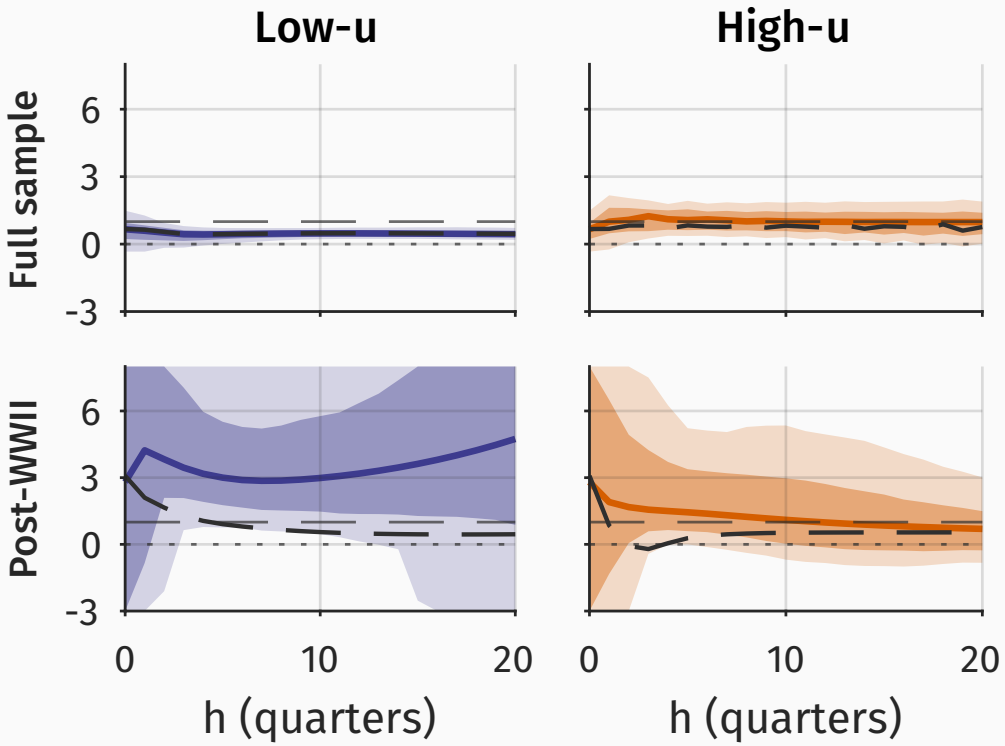
Post-WWII

low-u: 3.45 $\rightarrow$ 4.47

high-u: 1.56 $\rightarrow$ 0.72

Solid: estimated γ

Dashed: calibrated 6.5%



$\Rightarrow$ The direction **reverses** post-WWII, and only at the estimated threshold: there the regime definition binds

Why so wide? All samples

Normal-shock regime; multipliers from $h = 3$ to $h = 19$; 68% (dark) and 90% (light) bands

Multipliers at the Estimated Thresholds (2)

Cumulative multipliers at the estimated thresholds, normal-shock regime

	$h = 3$	$h = 7$	$h = 11$	$h = 19$
Full sample, $\hat{\gamma}^{\mu} = 8.42\%$				
low unemployment	0.44	0.46	0.48	0.45
high unemployment	1.24	1.06	0.99	0.98
Δ (high $-$ low)	**0.80**	0.59	0.51	0.53
90% interval for Δ	***[0.01, 1.41]***	*[-0.14, 1.45]*	*[-0.33, 1.47]*	*[-0.69, 1.62]*
Post-WWII, $\hat{\gamma}^{\mu} = 4.23\%$				
low unemployment	3.45	2.86	3.08	4.47
high unemployment	1.56	1.31	1.04	0.72
Δ (high $-$ low)	-1.89	-1.55	-2.03	-3.75
90% interval for Δ	*[-8.34, 3.86]*	*[-6.11, 2.99]*	*[-9.07, 4.14]*	*[-25.59, 13.77]*

- Full sample: larger in high-unemployment regime, significant at $h = 3$
- Post-WWII: larger in low-unemployment regime, in point estimates only

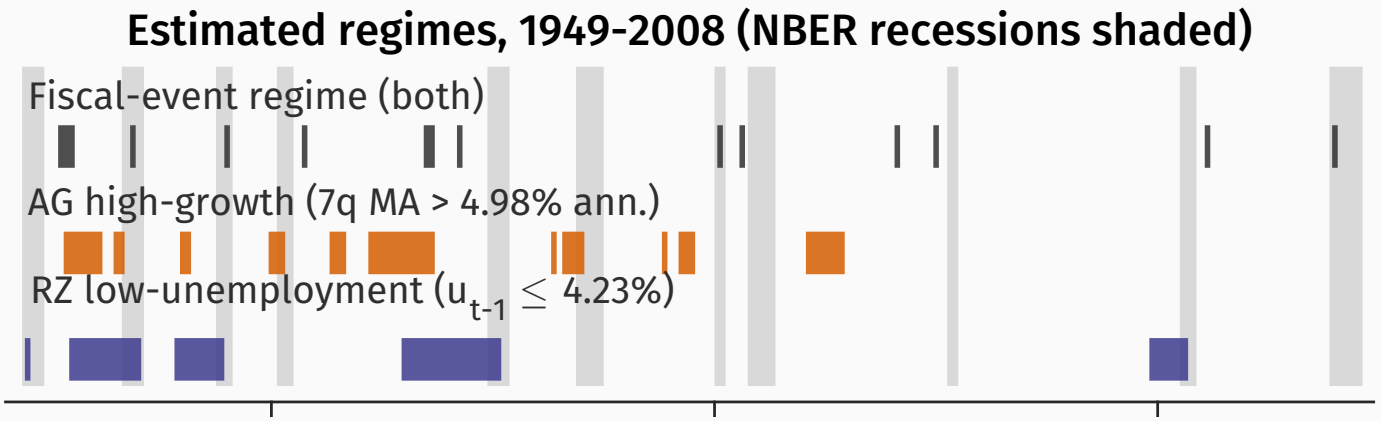
$\Rightarrow$ RZ2018: 0.60 vs 0.59. Here direction **not fixed: depends on the sample**

Is the Fiscal-Event Regime an Artifact of the Instrument?

Concern: the variance state $|z_{t-1}|$ is the instrument itself, and z_t is in the VAR

Robustness: AG2012 data, 1947Q1–2008Q4

- VAR in (G, T, Y), Blanchard–Perotti: **no instrument in the system**
- the news series is only a candidate variance state



⇒ Macro aggregates alone pick the **same fiscal-event quarters**: 15 of 15

⇒ Shock regimes: a feature of the economy, not of the instrument

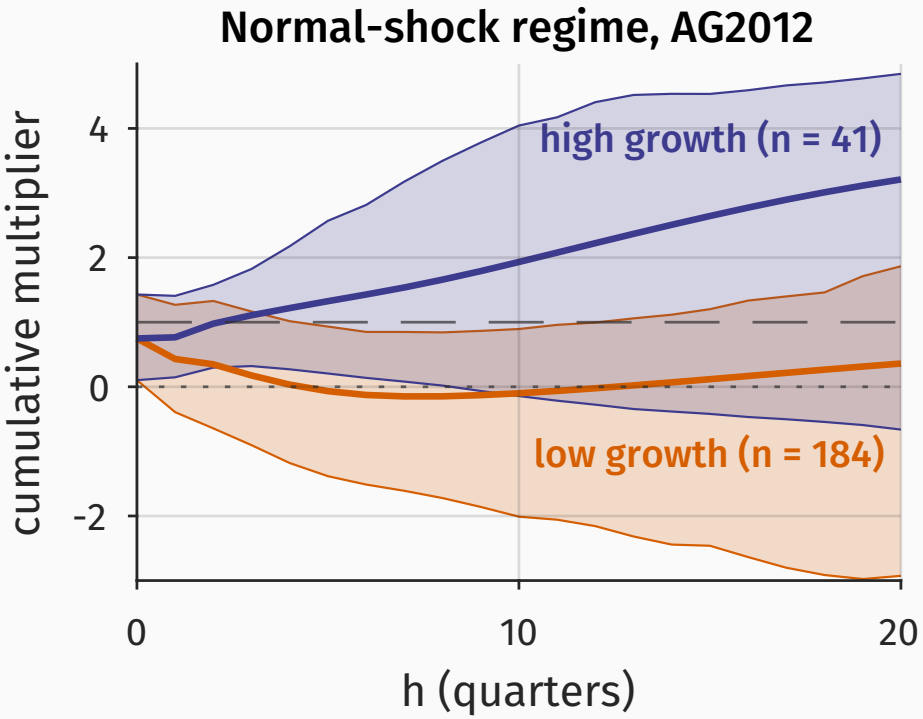
Post-WWII Multipliers Under a Different Identification

AG2012 data

Mean state: output growth over the past 7 quarters

- High-growth: 1.11 $\rightarrow$ 3.12
- Low-growth: 0.18 $\rightarrow$ 0.31
- Difference **significant through** $h = 11$

Tables



$\Rightarrow$ Post-WWII, multipliers are larger in good times: same direction as with RZ2018's data, now **significant**

Reconciling the Literature

Three ingredients account for the disagreement:

1. **Shock environment**
   - fiscal innovations far more volatile in few fiscal-event quarters; once isolated, state dependence appears at RZ2018 threshold
2. **Regime definition**
   - binds after WWII: the difference appears at the estimated 4.2%, not at 6.5%
3. **Sample period**
   - with pre-WWII data, extreme slack dominates: larger multipliers in high-unemployment regime, as in AG2012
   - post-WWII alone: larger in low-unemployment regime, as in Alloza2022

$\Rightarrow$ **A regime is two-dimensional**: a propagation regime (state of the economy) and a shock regime (shock environment)

Conclusion

Conclusion

1. **Model**: mean and variance with their own observable states and estimated thresholds; nests the single, common, variance-only and linear structures
2. **Estimator**: one criterion, the Gaussian quasi-likelihood, in two steps; thresholds at rate T and first-order costless: standard inference on regime parameters, IRFs, multipliers
3. **Application**: state dependence of spending multipliers appears once the fiscal-event quarters are isolated: larger in the high-unemployment regime with pre-WWII data, in the low-unemployment regime after WWII

Key takeaways:

1. Volatility regimes need a criterion that **prices the covariance**: the SSR estimator does not see them
2. Regimes are two-dimensional: ignoring the shock environment contaminates comparisons across states of the economy

Appendix

Leverage Points and the Full Sample

- At the calibrated threshold the full-sample gap is smaller than on the sample excluding WWII, although the fiscal-event regime is isolated in both
- Swapping regime coefficients and covariances across the two samples, one at a time: the whole difference comes from the coefficients of the high-unemployment regime
- Within those, from **three quarters**: 1941Q3–1942Q1, the U.S. entry into WWII
  - largest news realizations of the century, while unemployment is still high

  ⇒ extreme leverage points inside the regime

! Isolating the fiscal-event regime downweights these quarters in $\hat{\Phi}_i$ (GLS weights); excluding WWII removes them

Back

One Threshold: All Estimators

Single threshold on the state variable; regime 1 = below the threshold

	Likelihood		SSR (mean only)		Calibrated (RZ2018)	
Sample	$\hat{\gamma}$	regime 1 obs.	$\hat{\gamma}$	regime 1 obs.	γ	regime 1 obs.
Slack exercise (unemployment rate, %)						
Full sample	3.56	57 (11%)	6.95	343 (69%)	6.50	319 (64%)
Excl. WWII	3.64†	49 (10%)	4.20	105 (22%)	6.50	300 (63%)
Post-WWII	5.61	139 (51%)	5.61	139 (51%)	6.50	189 (70%)
Interest-rate exercise (3M T-bill rate, %)						
Full sample	3.48	223 (56%)	0.21‡	54 (14%)	0.25	58 (15%)
Excl. WWII	3.48	201 (53%)	1.22	94 (25%)	0.25	55 (15%)

† Smallest regime allowed by the trimming. ‡ Lowest feasible grid point.

- SSR (conditional mean only) reproduces RZ2018's cut-offs on the full sample: 6.95% and 0.21%
- The likelihood departs from them only because it also sees the second moments; post-WWII, with no war, the two coincide

Back

Assumption A4 at the Estimated Parameters

A4 restricts the DGP, but is checkable at the estimates: smallest λ with $\hat{A}_i' P \hat{A}_i \preceq \lambda^2 P$ in both regimes (SDP); a root ≥ 1 rules it out

Sample	Regime	Quarters	Largest root	Share < 1 (bootstrap)	A4
Full sample	$u_{t-1} \leq 8.42$	410	0.926	0.96	
	$u_{t-1} > 8.42$	89	**1.582**	0.00	fails
Excl. WWII	$u_{t-1} \leq 3.44$	41	0.963	0.59	holds, $\lambda^* = 0.985$
	$u_{t-1} > 3.44$	436	0.984	0.39	
Post-WWII	$u_{t-1} \leq 4.23$	52	1.045	0.24	fails
	$u_{t-1} > 4.23$	220	0.977	0.80	

Spectral radius of the estimated companion matrix; share over 499 bootstrap draws. Linear VAR: 0.91–0.97

- Post-WWII low-u: within sampling uncertainty (near-unit-root VAR in levels, 52 quarters)
- Full-sample high-u: a constant-coefficient law does not describe the Great Depression; outside the premise of Theorems 1–2, read at short horizons
- A4 is sufficient, not necessary. RZ2018's own TVAR has explosive roots under the slack state (fn. 42); at 6.5% the high-u regime is explosive in all samples

Back (theory) Back (multipliers)

Why Not Spending as the Variance State?

A shock regime should date the cause, not the response

- news is the identifying shock; spending is the response, quarters or years later

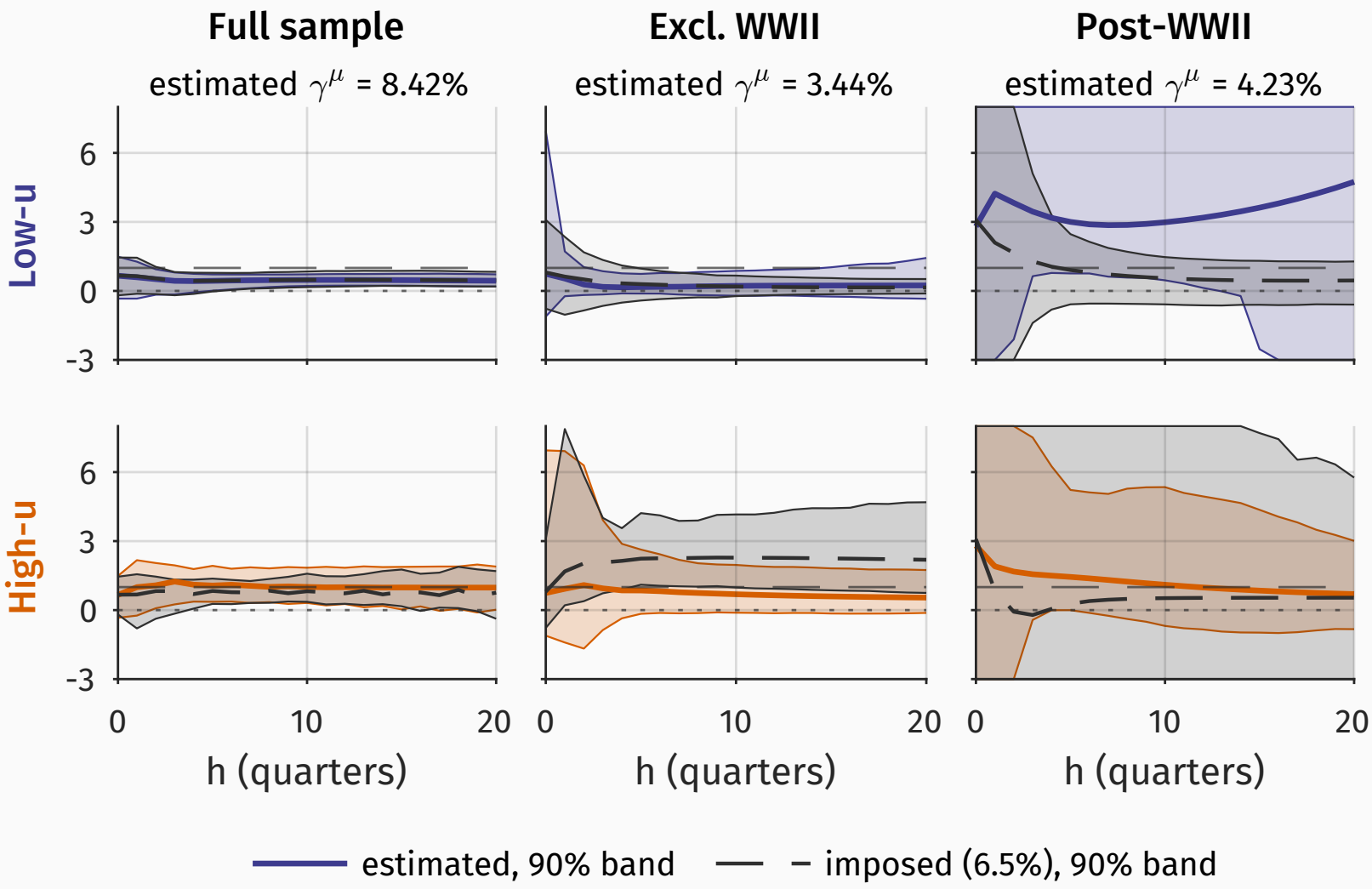
Check: same model with $|\Delta g_{t-1}|$ as the variance state

	Fiscal-event quarters		Agreement	Δ (high $-$ low), $h = 19$	
Sample	news	spending	(Cohen's κ)	news	spending
Full sample	33	92	0.16	0.53	0.06
Excl. WWII	25	71	0.12	spending: boundary partition	
Post-WWII	17	247	0.00	spending: boundary partition	

- Spending growth dates the build-ups **and** demobilizations of the major wars; it misses the announcement quarters and every fiscal event after 1968
- Full sample: mean threshold back at 3.56%, the one-threshold estimate

Back

Calibrated vs Estimated Threshold



Back

Is It the VAR or the Variance Regimes?

Same partition as RZ2018 (6.5%), three estimators (cumulative multipliers, high-u / low-u, $h = 7$)

	LP-IV (RZ2018)	VAR no variance regimes	VAR fiscal-event regime isolated
Full sample	0.60 / 0.59	0.52 / 0.60	0.77 / 0.46
Excl. WWII	0.72 / 0.56	1.66 / 0.42	2.26 / 0.24
Post-WWII	−1.63 / 0.80	1.25 / 0.54	0.45 / 0.72

- Full sample: the VAR at their partition **reproduces their null**; the difference comes from isolating the fiscal-event regime
- Excl. WWII: the VAR step itself moves the high-u multiplier, the one sample where RZ2018 report that the method matters (their fn. 26)
- Post-WWII: the VAR removes the negative multipliers; nothing at 6.5% is identified (79-quarter cell)

Back

How Strong Is the News Instrument?

First-stage HAC F of the news z_t for spending g_t, within each regime

	Low unemployment		High unemployment	
Threshold at 6.5%	all quarters	normal-shock	all quarters	normal-shock
Full sample	2.2	**11.6**	2.9	6.1
Excl. WWII	0.8	**13.7**	0.4	< 0.1
Post-WWII	1.5	3.1	0.6	1.3

- "All quarters" = RZ2018's regimes: weak, as they report
- Isolating the fiscal events **strengthens** the low-unemployment first stage: there the news–spending coefficient has the opposite sign
- Estimated thresholds (low/high): full sample 11.5/1.0, post-WWII 5.0/4.6
- No regime, RZ2018's included, reaches the Montiel Olea–Pflueger critical value (37.4 for a 5% bias)

! Excl. WWII, high unemployment: almost no news variation outside the fiscal events, as in RZ2018 ⇒ same caution for that multiplier

Back (setup) Back (results)

Are the Wide Bands the Bootstrap or the Object?

The object. The multiplier is a ratio; in two thin post-WWII cells the cumulative spending response changes sign in a non-trivial share of draws.

	cell (obs.)	$h = 3$	$h = 7$	$h = 11$	$h = 19$
Share of bootstrap draws in which the denominator crosses zero (%)					
Post-WWII, estimated	low unemployment (47)	3.0	2.0	3.2	8.8
Post-WWII, calibrated	high unemployment (79)	23.2	13.4	13.2	14.4
All other normal-shock cells	(84–415)			≤ 1.6	

- **Not the bootstrap:** a flat-prior Bayesian posterior (two-block Gibbs, thresholds plugged in) reproduces the bootstrap bands (widths 0.8–1.0× in the historical samples, 0.65–0.7× post-WWII)
- **Informative priors move results in both directions:** a standard Minnesota prior shrinks the proxy's propagation to zero (news $\to y$: 0.10 $\to$ 0.005); freeing the proxy column restores the frequentist picture; long-horizon ratios in regimes that are explosive in-sample stay fragile under any prior

$\Rightarrow$ Bootstrap kept; figures show 68% and 90% bands, tables 90%; the calibrated-threshold result excl. WWII survives every treatment

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Multipliers at the Estimated Thresholds: All Samples

Stability at the estimates Back

Interest-Rate Exercise: Thresholds

State: lagged 3-month T-bill rate; RZ2018 calibrate the threshold at 0.25%

	One threshold		Decoupled	Calibrated
Sample	likelihood	SSR	$\hat{\gamma}^{\mu}$	(RZ2018)
Full sample	3.48	0.21	0.24	0.25
Excl. WWII	3.48	1.22	2.73	0.25

- One threshold, likelihood: a broad low-rate environment, not a zero lower bound
- With the fiscal-event regime isolated ($\hat{\gamma}^{\sigma} \approx 0.024$ again), the full-sample mean threshold **recovers RZ2018's** 0.25%
- Calibrated threshold, excl. WWII: low-rate 2.5–2.8 vs high-rate 0.6, significant (50 low-rate quarters: the 1930s and post-2008)

Back

Interest-Rate Exercise: Calibrated Threshold

- Mean threshold fixed at the calibrated 0.25%, variance threshold estimated; normal-shock regime, 68% (dark) and 90% (light) bands

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Interest-Rate Exercise: Estimated Thresholds

Back

AG2012 Exercise: Thresholds and Multipliers

	Thresholds		Low growth, obs.		High growth, obs.	
Variance state	$\hat{\gamma}^{\mu}$ (%)	$\hat{\gamma}^{\sigma}$	normal	fiscal-event	normal	fiscal-event
$\lvert \text{news}_{t-1} \rvert$ (% of GDP)	1.245	2.036	184	11	41	4

Normal-shock multipliers	$h = 3$	$h = 7$	$h = 11$	$h = 19$
low growth	0.18	−0.15	−0.06	0.31
high growth	1.11	1.54	2.08	3.12
Δ (high − low)	**0.93**	**1.69**	**2.14**	2.80
90% interval	*[0.05, 1.79]*	*[0.15, 3.76]*	*[0.06, 5.04]*	*[-0.56, 5.89]*

- $\hat{\gamma}^{\mu}$: on the backward 7-quarter average of output growth ($\approx 5\%$ annualized); AG2012's centered average uses future growth
- High-growth and RZ2018's low-unemployment regime share only 16 quarters: same direction under both definitions of "good times"

Back

Monte Carlo: One-Threshold Designs

RMSE of the SSR and likelihood (joint) estimators

		Threshold $\hat{\gamma}$		$\hat{\Phi}_i$		$\hat{\Sigma}_i$	
Break in	Estimator	$T = 200$	$T = 1000$	$T = 200$	$T = 1000$	$T = 200$	$T = 1000$
none	SSR	–	–	1.758	2.259	0.360	0.263
	likelihood	–	–	1.908	2.590	0.495	0.426
mean only	SSR	0.025	0.004	0.334	0.147	0.246	0.109
	likelihood	0.025	0.004	0.334	0.147	0.247	0.109
variance only	SSR	2.004	3.204	2.902	4.138	1.756	1.629
	likelihood	0.080	0.013	0.454	0.191	0.760	0.337
mean and variance	SSR	0.081	0.010	0.469	0.197	0.721	0.315
	likelihood	0.039	0.006	0.451	0.197	0.712	0.315

- No regime in the data: the estimated split selects small tail regimes, parameter errors grow with $T \Rightarrow$ test for a threshold first (Hansen 1996)

Back

Monte Carlo: Nesting the Joint Model

Unrestricted estimator with a shared state: RMSE and selection of the joint model

		Thresholds		$\hat{\Phi}_i$		Joint model selected	
True model	T	$\hat{\gamma}^{\mu}$	$\hat{\gamma}^{\sigma}$	unrestr.	joint	$c_T = \ln T$	$c_T = 2 \ln T$
joint, mean break only	200	0.024	1.712	0.339	0.334	0.813	0.990
	1000	0.005	1.959	0.148	0.147	0.946	0.998
joint, variance break only	200	1.954	0.092	3.788	0.454	0.800	0.997
	1000	2.894	0.013	5.127	0.191	0.920	1.000
joint, both break	200	0.050	0.078	0.464	0.451	0.992	1.000
	1000	0.010	0.014	0.198	0.197	0.999	1.000
decoupled (two designs)	200					0.000	0.000
	1000					0.000	0.000

- Both moments break: both thresholds at rate T, $\hat{\Phi}_i$ as precise as with the joint estimator
- Only one breaks: the other threshold is not identified; the joint model is still selected

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Monte Carlo: Limiting Distributions

Thresholds: Kolmogorov–Smirnov distance between $T(\hat{\gamma} - \gamma^0)$ and the simulated limit

	joint $\hat{\gamma}$	$\hat{\gamma}^\mu$	$\hat{\gamma}^\sigma$	5% critical value
$T = 200$	0.011	0.009	0.031	0.019
$T = 1000$	0.024	0.025	0.012	0.030

Regime parameters and independence

- empirical over asymptotic standard deviations: 1.04–1.05 at $T = 200$, 1.00 at $T = 1000$
- correlation between the two normalized thresholds: -0.007 and 0.013

Plug-in confidence intervals for the variance threshold

- coverage at nominal 90% / 95%: 0.86 / 0.92 at $T = 200$, 0.89 / 0.94 at $T = 1000$, with true or estimated covariances

Back

Joint (Common-Threshold) Model and Estimator

- Let $y_t \in \mathbb{R}^n$ follow a threshold-VAR(p) (TVAR) with k regimes and changing conditional variances

$$y_t = \sum_{i=1}^{k} \left(\Phi_{i,0} + \sum_{j=1}^{p} \Phi_{i,j} y_{t-j} + \Sigma_i^{1/2} \varepsilon_t \right) \mathbb{I}(s_t \in \mathcal{R}_i), \quad \varepsilon_t \sim NID(0, I_n)$$

- **SET-VAR model:** $s_t = e_s' y_{t-d} = q_{t-d}$ with $d \in \{1, \ldots, p\}$

Likelihood-based Estimation:

- Define $\theta = \{\gamma', \phi_1', \ldots, \phi_k', \text{vech}(\Sigma_1)', \ldots, \text{vech}(\Sigma_k)'\}$
- y_t, conditional on $\mathcal{F}_{t-1}$, given regime i is Gaussian

$\Rightarrow$ Objective function: Gaussian piecewise linear average log-likelihood $\ell_T(\theta)$

- The estimator is

Estimation Algorithm

$$\hat{\theta}_T = \arg\max_{\theta \in \Theta} \ell_T(\theta), \quad \text{and} \quad \hat{\gamma} = \arg\max_{\gamma \in \Gamma} \ell_T(\gamma, \phi(\gamma), \Sigma(\gamma))$$

Joint Model: Theoretical Results (1)

1. Geometric ergodicity and finite moments:

   Geometric ergodicity definition

   $\Rightarrow$ **Theorem 3 (i):** under A1-A4, geometric ergodicity and finite fourth moments

   Assumptions A | Theorem 3 (i)

2. Proper parameter identification:
   - Condition for regime separation: discontinuity in likelihood
   - Regime-specific parameters need to differ

   Assumptions B1–B4

- **Consistency:** under the assumptions above

$$\hat{\theta}_T \to_{a.s.} \theta_0 \qquad \text{as} \quad T \to \infty$$

Theorem 3 (ii)

Joint Model: Theoretical Results (2)

3. Regularity local to the threshold: where the limit of $\hat{\gamma}$ is decided
   - Weak dependence, conditional moments and continuity near γ^0

   Assumptions B5–B7

- **Convergence rates:** under the assumptions above
  - (i) Threshold is super-consistent: $T(\hat{\gamma} - \gamma^0) = O_p(1)$
  - (ii) Standard $\sqrt{T}$-consistency of the regime parameters

  Theorem 3 (iii)

- **Asymptotic distribution:** under the assumptions above
  - (i) Threshold: compound-Poisson argmax limit at rate T (fixed-threshold-effect framework, as Chan 1993)
  - (ii) Regime parameters: Gaussian, as if γ^0 were known, asymptotically independent of $\hat{\gamma}$

  Theorem 3 (iv)

  - ! Limit law of $\hat{\gamma}$ not pivotal: bootstrap invalid

  Inference

Joint Model: Estimation Algorithm

1. **Grid search over Γ**: for each $\gamma^{(j)} \in \Gamma$
   - Assign observations: $t \in \mathcal{R}_i\left(\gamma^{(j)}\right)$ iff $\gamma_{i-1}^{(j)} < y_{t-d} \leq \gamma_i^{(j)}$
   - Compute *closed-form* conditional MLEs:

$$\hat{\phi}_i\left(\gamma^{(j)}\right) = \left(\sum_t \mathbb{I}_{it}\, x_t x_t'\right)^{-1}\left(\sum_t \mathbb{I}_{it}\, x_t y_t'\right) \quad \text{(OLS within regime)}$$

$$\hat{\Sigma}_i\left(\gamma^{(j)}\right) = \frac{1}{n_i}\sum_t \mathbb{I}_{it}\, \hat{\varepsilon}_{it}\hat{\varepsilon}_{it}', \qquad \hat{\varepsilon}_{it} = y_t - \left(x_t' \otimes I_n\right)\hat{\phi}_i$$

   - Evaluate the **profiled log-likelihood**:

$$\ell_T^P\left(\gamma^{(j)}\right) = \frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{k}\left[-\tfrac{1}{2}\log\det\hat{\Sigma}_i - \tfrac{1}{2}\hat{\varepsilon}_{it}'\hat{\Sigma}_i^{-1}\hat{\varepsilon}_{it}\right]\mathbb{I}\left(y_{t-d} \in \mathcal{R}_i\left(\gamma^{(j)}\right)\right)$$

$$\hat{\gamma} = \arg\max_{\gamma^{(j)} \in \Gamma}\ \ell_T^P\left(\gamma^{(j)}\right)$$

2. **Conditional on $\hat{\gamma}$**: set $\hat{\phi}_i = \hat{\phi}_i(\hat{\gamma})$ and $\hat{\Sigma}_i = \hat{\Sigma}_i(\hat{\gamma})$, $i = 1, \ldots, k$

$\Rightarrow$ Super-consistency of $\hat{\gamma}$: estimation uncertainty in Step 1 has no first-order effect on $\hat{\phi}_i$, $\hat{\Sigma}_i$ (Theorem 3) Back

Geometric Ergodicity

The Markov chain $\{X_t\}$ is geometrically ergodic if it has a unique invariant probability measure π and there exists constants $M(x) < \infty$ and $\rho \in (0,1)$ such that for all x and integers $t \geq 0$

$$\|P^t(x,\cdot) - \pi(\cdot)\|_{TV} \leq M(x)\rho^t$$

Equivalently, there exists a function $g \geq 1$ and constants $R < \infty$ and $\rho \in (0,1)$ with $\|P^t(x,\cdot) - \pi(\cdot)\|_{TV} \leq Rg(x)\rho^t$; in particular geometric ergodicity implies geometric β-mixing and exponential forgetting of initial conditions.

Back (joint) Back (decoupled)

Assumptions for Geometric Ergodicity

A1 The sequence $\{\varepsilon_t\}$ is i.i.d.$(0, I_n)$, everywhere positive continuous density on $\mathbb{R}^n$, finite fourth moments.

A2 For each i the matrices $\Phi_{i,j}$ and $\Phi_{i,0}$ are finite constants; the regime-selection map $i^*(x)$ is measurable, so that $X_t = (y_t, y_{t-1}, \ldots, y_{t-p+1})'$ is a time-homogeneous Markov chain.

A3 Each regime volatility matrix Σ_i is non singular, and $\sup_{1 \le i \le k} \|\Sigma_i\| = C_\Sigma < \infty$.

A4 There exists a P.D. matrix $P \succ 0$ and constants $\lambda_i \in (0, 1)$ s.t.

$$A_i' P A_i \preceq \lambda_i^2 P, \qquad \forall i \in 1, \ldots, k$$

A_i: companion matrix of regime i; a common quadratic Lyapunov function, not just $\rho(A_i) < 1$ regime by regime

Back (joint) Back (decoupled)

Joint Model: Assumptions for Identification

B1 The parameter space $\Theta = \Gamma \times \mathcal{B} \times \mathcal{S}$ is compact:

(i) $\Gamma = \{\gamma \in \mathbb{R}^{k-1} | \underline{\gamma} \leq \gamma_1 \leq \cdots \leq \gamma_{k-1} \leq \overline{\gamma}\}$ for finite $\underline{\gamma}$, $\overline{\gamma}$

(ii) $\mathcal{B} = \prod_{i=1}^{k} \mathcal{B}_i$, with each $\mathcal{B}_i$ compact and ensures stability (A4 guarantees $\rho(A_i) < 1$)

(iii) $\mathcal{S} = \prod_{i=1}^{k} \mathcal{S}_i$, where $\mathcal{S}_i = \{\Sigma \in \mathbb{R}^{n \times n} | \underline{\sigma} I \prec \Sigma \prec \overline{\sigma} I\}$ for $0 < \underline{\sigma} < \overline{\sigma} < \infty$

B2 For each $i = 1, \ldots, k$ the true threshold parameter γ^0 satisfies:

(i) Positive density at thresholds: marginal density f_π of state y_{t-d} under the stationary measure π is s.t. $f_\pi(\gamma^0) > 0$

(ii) Regime separation: for every sufficiently small neighborhood $\mathcal{N}_i$ of γ_i^0, $\exists\, A_i \subseteq \{y_{t-d} \in \mathcal{N}_i\}$ with $\pi(A_i) > 0$ such that the adjacent conditional Gaussian laws are distinct on A_i, i.e.

$$\left((x_t' \otimes I_n)\mathrm{vec}(B_i^0), (\Sigma_i^0)\right) \neq \left((x_t' \otimes I_n)\mathrm{vec}(B_{i+1}^0), (\Sigma_{i+1}^0)\right) \quad \text{on } A_i.$$

Equivalently, on a set of positive stationary probability in any neighborhood of γ_i^0, either the conditional mean, or the cond. covariance matrix differ across adjacent regimes (or both).

B3 For $i \neq j$, $(\Phi_i, \Sigma_i) \neq (\Phi_j, \Sigma_j)$

B4 The true parameter $\theta_0 \in \Theta$

Back

Joint Model: Geometric Ergodicity (Theorem 3(i))

Theorem 3 (i). *Under assumptions A1 - A4, the Markov chain $\{X_t\}$ is geometrically ergodic. Moreover, the unique invariant probability measure π satisfies*

$$\int \|x\|^4 \pi(dx) < \infty$$

so in particular the VAR has finite fourth moments under stationarity.

Back

Joint Model: Proof Sketch, Geometric Ergodicity

Results used:

Lemma 1. *Under assumptions A1 to A3, the Markov chain (X_t) defined by the model is aperiodic and ν-irreducible with ν being the Lebesgue measure. Furthermore, a ν-non-null compact set is a small set.*

Lemma 2. *Suppose that (X_t) is an aperiodic and ν-irreducible, where ν is the Lebesgue measure. If there exists a small set A, a non-negative measurable function g and constants $\rho \in (0,1)$ and $b > 0$ such that:*

(i) *g is bounded away from zero and infinity on A;*

(ii) $\sup_{x \in A} \mathbb{E}\{g(X_t)|X_{t-1} = x\} < \infty$;

(iii) $\mathbb{E}\{g(X_t)|X_{t-1} = x\} < \rho g(x) - b, x \in A^c$;

then (X_t) is geometrically ergodic and

$$\int_{\mathbb{R}^{np}} g(x)\pi(dx) < \infty \tag{1}$$

where π is an invariant probability measure on $\mathbb{R}^{np}$ with respect to the Markov chain (X_t).
(Tong (1990), Meyn and Tweedie (1993))

$\Rightarrow$ **Proof idea:** Lemma 1 gives irreducibility and aperiodicity; verify the drift conditions of Lemma 2 twice, for a quadratic $g(x) = 1 + x'Px$ (geometric ergodicity) and for a quartic G (fourth moments)

Joint Model: Proof Sketch, Geometric Ergodicity (2)

- For each regime i, define the lifted matrices $B_i = A_i \otimes A_i \in \mathbb{R}^{(np)^2 \times (np)^2}$
- Under A4 $A_i' P A_i \preceq \lambda_i^2 P$ for $P \succ 0$, $\lambda_i \in (0,1)$, then $B_i'(P \otimes P)B_i \preceq \lambda_i^4 (P \otimes P)$

  $\Rightarrow$ Stable lifted system with rate $\lambda^{(4)} = \max_{1 \le i \le k} \lambda_i^4 < 1$
- Lemma 1 applies under A1 - A3, only **NTS** one-step drift inequality and apply lemma 2
- Quartic Lyapunov function: $G(x) := 1 + (x \otimes x)' Q (x \otimes x) \;=\; 1 + Z' Q Z$, where Q satisfies lifted A4: $B_i' Q B_i \preceq \lambda^{(4)} Q, \qquad \forall i = 1, \ldots, k$
- **Goal:** bound $\mathbb{E}\big[G(X_{t+1}) \mid X_t = x\big] \;=\; 1 + \mathbb{E}\big[Z_{t+1}' Q Z_{t+1} \mid X_t = x\big]$
- Remainder terms bounded using Young's inequality, pure noise terms bounded by $\|\varepsilon\|^4 < \infty$
- Lemma 2 (iii) proven (drift outside a compact set)

Joint Model: Proof Sketch, Geometric Ergodicity (3)

- Lemma 2 (i)-(ii) define a compact set $A = \{x : \|x\| \leq R\}$ (small)
- Using G, B_i and other terms being bounded on $A \Rightarrow$ $\sup_{x \in A} \mathbb{E}[G(X_{t+1})|X_t = x] < \infty$ (lemma 2 (ii) proven)
- Lemma 2 (i):
  - $G(x) \geq 1$ for all x
  - On compact set A, $G(x) < \infty$, using bounds from (ii)

Lemma 2 applies and

1. Yields geometric ergodicity and $\int G \, d\pi < \infty$
2. Since $G(x) = 1 + (x \otimes x)' Q (x \otimes x)$ controls $\|x\|^4$ (by equivalence of norms), we also have $\int \|x\|^4 \, \pi(dx) < \infty$
3. The finite fourth moment of X_t implies that $\mathbb{E}\|y_t\|^4 < \infty$, under stationarity.

Back

Joint Model: Consistency and Rates (Theorem 3(ii)–(iii))

Theorem 3 (ii). *Under assumptions A1 - A4 (geometric ergodicity and finite fourth moments) and B1 - B4 (identification), if the data $\{y_t\}_{t=1}^{T}$ is generated by the true parameter θ_0:*

$$\hat{\theta}_T \to_{a.s.} \theta_0 \qquad \text{as} \quad T \to \infty$$

Theorem 3 (iii). *Under assumptions A1 - A4 and B1 - B7,*

(i) *Super consistency of the threshold*

$$T(\hat{\gamma} - \gamma^0) = O_p(1)$$

(ii) *$\sqrt{T}$-consistency of the regime parameters: for each $i = 1, \ldots, k$*

$$\sqrt{T} \begin{pmatrix} \hat{\Phi}_i - \Phi_i^0 \\ \text{vech}(\hat{\Sigma}_i - \Sigma_i^0) \end{pmatrix} = O_p(1)$$

Proof Sketches Back

Joint Model: Proof Sketch, Consistency

- Key inputs:
  1. **Ergodicity and moments** (Theorem 3(i)): gives geometric β-mixing and $\mathbb{E}_\pi \|X_t\|^4 < \infty$
  2. **ULLN**: geometric mixing + integrable envelope $\Rightarrow$ ULLN for log-likelihood over Θ

- **Identification**: if $\gamma \neq \gamma^0$, the regime partition differs on a set of positive stationary probability (B2); regime-specific Σ_j help ensure conditional densities differ $\Rightarrow$ KL divergence > 0

- **Conclusion**: uniform convergence + identification $\Rightarrow$ argmax theorem (Newey and McFadden 1994) gives $\hat{\theta} \to \theta_0$

Joint Model: Proof Sketch, Rates

The localization argument of Chan (1993), Hansen (2000) and Samia and Chan (2011), for a multivariate likelihood with regime-specific covariances

- **Local process**: $\mathcal{L}_T(h) = T\left[\ell_T^P\left(\gamma^0 + \frac{h}{T}\right) - \ell_T^P(\gamma^0)\right]$; show its maximizer is $O_p(1)$
- **Local counting**: the number of observations with q_{t-d} between γ^0 and $\gamma^0 + h/T$ has expectation $\approx f(\gamma^0)|h|$ (B2, stationarity) $\Rightarrow$ only $O_p(1)$ observations change regime at scale $1/T$
- **Loss per switched observation**: evaluated under the wrong regime law, each pays a strictly positive expected KL loss (B2–B3)
- **Remainder**: the other observations enter only through the re-estimated regime parameters, which move by $O_p(1/T)$ (Lemma 7, profile stability) $\Rightarrow o_p(1)$ after scaling by T

$\Rightarrow$ for $|h|$ large the local process is negative w.p. $\to 1$: $T(\hat{\gamma} - \gamma^0) = O_p(1)$

Joint Model: Proof Sketch, Rates (2)

- **Negligible misclassification**: since $|\hat{\gamma} - \gamma^0| = O_p(1/T)$, number of misclassified obs. is $O_p(1)$ (finite sample nuisance!)
- **Implication**: threshold super consistent $\Rightarrow$ asymptotically no first-order contamination in the $\sqrt{T}$ regime parameter estimation

$\Rightarrow$ Given negligible misclassification, mixing and finite 4th moments, use standard MLE asymptotics (as Hansen 2000)

Back

Joint Model: Assumptions for the Limiting Distribution

Envelope: $\Lambda_t := 1 + \|x_t\|^2 + \|y_t\|^2$; all conditions local to γ^0, where the limit law of $\hat{\gamma}$ is decided

B5 (Local ρ-mixing) $\exists\, \Delta > 0$ such that
$W_t := \left(\Lambda_t \mathbb{I}\{|q_{t-d} - \gamma^0| \le \Delta\},\ q_{t-d}\, \mathbb{I}\{|q_{t-d} - \gamma^0| \le \Delta\}\right)'$ is ρ-mixing with summable mixing coefficients

- Controls dependence near the threshold (Samia–Chan 2011); dispensable here: implied by ergodicity + strengthened B6

B6 (Conditional moments near threshold) $\exists\, \tau > 0, M < \infty$ such that
$\mathbb{E}\left[\Lambda_t^2 \mid q_{t-d} = v\right] \le M$ for a.e. $v \in [\gamma^0 - \tau, \gamma^0 + \tau]$

- Likelihood jumps dominated by $C\Lambda_t \Rightarrow$ tightness (not implied by finite 4th moments)

B7 (Weak continuity at threshold) The conditional law of $x_t^{(-)}$ given $q_{t-d} = v$ is weakly continuous at $v = \gamma^0$

- Makes the one-sided limit jump laws $\mu^{\pm}$ well-defined; not needed for ε_t (independent of $\mathcal{F}_{t-1}$)

Back

Joint Model: Limiting Distributions (Theorem 3(iv))

Jump variables: log-likelihood loss from evaluating an obs. under the wrong regime law, e.g. a true-regime-2 obs. under regime 1:

$$\xi_t^+ := m_1(y_t) - m_2(y_t), \qquad \mathbb{E}\left[\xi_t^+ | x_t\right] = -\,\mathrm{KL}\left(N(B_2^0 x_t, \Sigma_2^0) \,\|\, N(B_1^0 x_t, \Sigma_1^0)\right) \leq 0$$

with one-sided limit laws $\mu^{\pm}$ at the threshold (well-defined under B6, B7)

Theorem 3 (iv). *Let $k = 2$. Under A1–A4 and B1–B7:*

(i) *The local process $T\left[\ell_T^P\left(\gamma^0 + \frac{\kappa}{T}\right) - \ell_T^P(\gamma^0)\right]$ converges weakly to a two-sided compound Poisson process with rate $f_\pi(\gamma^0)$ and jump laws $\mu^{\pm}$*

(ii) *$T(\hat{\gamma} - \gamma^0) \Rightarrow M_-$, left endpoint of the a.s.-unique argmax interval*

(iii) *Asymptotically independent of $\hat{\gamma}$, as if γ^0 were known:*

$$\sqrt{T}\,\mathrm{vec}\left(\hat{\Phi}_i - \Phi_i^0\right) \Rightarrow N\left(0,\, S_i^{-1} \otimes \Sigma_i^0\right)$$

$$\sqrt{T}\,\mathrm{vech}\left(\hat{\Sigma}_i - \Sigma_i^0\right) \Rightarrow N\left(0,\, \tfrac{2}{\pi_i} D_n^+ \left(\Sigma_i^0 \otimes \Sigma_i^0\right) D_n^{+\prime}\right)$$

with all Gaussian blocks mutually independent

$S_i = \mathbb{E}_\pi\left[x_t x_t' \mathbb{I}\{q_{t-d} \in \mathcal{R}_i^0\}\right]$, $\pi_i = P_\pi(q_{t-d} \in \mathcal{R}_i^0)$, D_n^+: MP-inverse of duplication matrix. **General k:** the $k-1$ thresholds converge to *mutually independent* limits M_i.

Proof Sketch Back

Joint Model: Proof Sketch, Limiting Distributions

- **Local process**: moving γ by κ/T reassigns only the $O_p(1)$ obs. with q_{t-d} between γ^0 and $\gamma^0 + \kappa/T$; each contributes a jump $\xi_t^{\pm}$
- **Point-process convergence**: boundary obs. arrive as Poisson with rate $f_\pi(\gamma^0)$ (local counting + B5); jump marks converge to $\mu^{\pm}$ (B6 tightness, B7 continuity) $\Rightarrow$ compound Poisson limit in $D[0,\infty) \times D[0,\infty)$
- **Argmax**: jump means $= -\mathrm{KL} < 0$ (B2) $\Rightarrow$ both random walks drift to $-\infty$ $\Rightarrow$ a.s. unique maximizing interval $[M_-, M_+)$; argmax continuous mapping $\Rightarrow T(\hat{\gamma} - \gamma^0) \Rightarrow M_-$
- **Identification via 2nd moment**: with equal means, KL $= \frac{1}{2}\sum_j(\lambda_j - \ln\lambda_j - 1)$, $\lambda_j = \mathbf{eig}(\Sigma_1^{0,-1}\Sigma_2^0)$, > 0 iff $\Sigma_1^0 \neq \Sigma_2^0 \Rightarrow$ variance shifts alone localize γ at rate T (population counterpart of the SSR failure in the MC)
- **Regime parameters**: misclassified obs. $O_p(1) \Rightarrow$ no first-order contamination; standard MLE CLT within regimes, independence across blocks from the information structure

Back

Inference on the Thresholds

- Limit law M_- is not pivotal: depends on $f_\pi(\gamma^0)$ and the jump laws $\mu^\pm$
- ! Nonparametric bootstrap is inconsistent for $\hat{\gamma}$ under fixed-threshold-effect asymptotics (Yu 2014)
  - Bootstrap bands for objects computed *conditional* on $\hat{\gamma}$ (e.g. regime-specific IRFs) remain justified, by super-consistency
- Valid routes: subsampling (Gonzalo and Wolf 2005); simulation plug-in of estimated jump laws into the compound-Poisson limit (Li and Ling 2012)
- **Decoupled model, variance threshold:** jump law (V2) depends only on $\mathbf{eig}(\Sigma_1^{0,-1}\Sigma_2^0)$ — no conditional feature of the state at the threshold

⇒ Limit law of $T(\hat{\gamma}^\sigma - \gamma^{\sigma,0})$ fully determined by $(f_\sigma(\gamma^{\sigma,0}), \Sigma_1^0, \Sigma_2^0)$: near-pivotal

⇒ Fully parametric plug-in CIs: estimate $\hat{\lambda}_j$ from $(\hat{\Sigma}_1, \hat{\Sigma}_2)$ and $\hat{f}_\sigma(\hat{\gamma}^\sigma)$ by kernel, simulate the compound-Poisson argmax, read off quantiles

- No analogous shortcut for $\hat{\gamma}^\mu$ or the joint model's $\hat{\gamma}$

Back (joint) Back (decoupled)

Estimation Algorithm

1. **2D grid search** over $\Gamma^{\mu} \times \Gamma^{\sigma}$: for each $\left(\gamma^{\mu,(j)}, \gamma^{\sigma,(\ell)}\right)$
   - **Assign** each obs. to mean-regime i via $q^{\mu}_{t-d_{\mu}} \in \mathcal{R}^{\mu}_{i}(\gamma^{\mu,(j)})$ and variance-regime ℓ via $q^{\sigma}_{t-d_{\sigma}} \in \mathcal{R}^{\sigma}_{\ell}(\gamma^{\sigma,(\ell)})$; define joint cells $\mathcal{S}_{i\ell}$
   - **Preliminary OLS** within mean-regime i, ignoring heteroskedasticity $\Rightarrow$ residuals $\hat{u}_{it}$; pool within variance-regime ℓ:

$$\hat{\Sigma}^{(0)}_{\ell} = \frac{1}{t_{\ell}} \sum_{i} \sum_{t \in \mathcal{S}_{i\ell}} \hat{u}_{it}\hat{u}'_{it}, \qquad t_{\ell} = \sum_{i} t_{i\ell}$$

   - **Feasible GLS** for $\hat{B}_i$: solve normal eqs. using $\left(\hat{\Sigma}^{(0)}_{\ell}\right)^{-1}$ precision weights; compute GLS residuals $\hat{\varepsilon}_{it} = y_t - (x'_t \otimes I_n)\hat{\beta}^{\mathrm{GLS}}_{i}$; update pooled variance-regime covariance:

$$\hat{\Sigma}_{\ell} = \frac{1}{t_{\ell}} \sum_{i} \sum_{t \in \mathcal{S}_{i\ell}} \hat{\varepsilon}_{it}\hat{\varepsilon}'_{it}$$

   - **Evaluate** profiled log-likelihood — *single sum over variance regimes* ℓ:

$$L^{P}_{T}\left(\gamma^{\mu,(j)}, \gamma^{\sigma,(\ell)}\right) = -\frac{1}{2} \sum_{\ell=1}^{k_{\sigma}} \frac{t_{\ell}}{T} \ln \det \hat{\Sigma}_{\ell} + \mathrm{const}, \quad (\hat{\gamma}^{\mu}, \hat{\gamma}^{\sigma}) = \arg \max_{\Gamma^{\mu} \times \Gamma^{\sigma}} L^{P}_{T}$$

Estimation Algorithm (2)

2. Conditional on $(\hat{\gamma}^{\mu}, \hat{\gamma}^{\sigma})$: set $\hat{B}_i = \hat{B}_i^{\mathrm{GLS}}(\hat{\gamma}^{\mu}, \hat{\gamma}^{\sigma})$ and $\hat{\Sigma}_\ell = \hat{\Sigma}_\ell(\hat{\gamma}^{\mu}, \hat{\gamma}^{\sigma})$, $i = 1, \ldots, k_\mu$, $\ell = 1, \ldots, k_\sigma$

$\Rightarrow$ **Key difference from joint model:** profile likelihood sums over *variance regimes* ℓ only; $\hat{B}_i$ estimated by **GLS** (not OLS), using variance-regime precision weights

$\Rightarrow$ Super-consistency of $\hat{\gamma}^{\mu}$ and $\hat{\gamma}^{\sigma}$ (Theorem 2)

Back

Theorem A1: Geometric Ergodicity

Theorem A1. *Under assumptions A1–A4, the Markov chain $\{X_t\}$ is geometrically ergodic: it has a unique invariant probability measure π and, with $g(x) = 1 + x'Px$ for the matrix P of A4,*

$$\|P^t(x, \cdot) - \pi\|_{TV} \leq R\, g(x)\, \rho^t \qquad \text{for all } x,\ t \geq 1,$$

for some $R < \infty$, $\rho \in (0, 1)$. Moreover $\int \|x\|^4 \pi(dx) < \infty$: the VAR has finite fourth moments under stationarity.

- A4 restricts the k_μ **mean** companion matrices only; the Σ_ℓ enter through a bounded noise loading
- Gaussianity is not used here: A1 asks only for i.i.d. innovations with a positive continuous density and finite fourth moments
- Proof: as in the joint model (Lemmas 1–2, quadratic and quartic Lyapunov functions)

Definition Proof sketch

Back

Assumptions for Identification

B1 (Compact parameter space) $\Theta = \Gamma^{\mu} \times \Gamma^{\sigma} \times \mathcal{B} \times \mathcal{S}$ compact: ordered thresholds in $[\underline{\gamma}, \overline{\gamma}]$; $\mathcal{B}$ compact and consistent with A4; $\underline{\sigma} I \prec \Sigma_{\ell} \prec \overline{\sigma} I$

B2 (Separation at the thresholds) Mean side: f_{μ} continuous and positive at $\gamma_i^{\mu,0}$, and $P\big((B_{i+1}^0 - B_i^0)x_t \neq 0 \mid q_{t-d_{\mu}}^{\mu} = \gamma_i^{\mu,0}\big) > 0$. Variance side: f_{σ} continuous and positive at $\gamma_{\ell}^{\sigma,0}$, and $\Sigma_{\ell}^0 \neq \Sigma_{\ell+1}^0$. Every feasible cell has $\pi_{i\ell} > 0$

B3 (Regimes are different) $B_i^0 \neq B_{i'}^0$ for $i \neq i'$; $\Sigma_{\ell}^0 \neq \Sigma_{\ell'}^0$ for $\ell \neq \ell'$

B4 (Interior) θ_0 is interior to Θ

B9 (Candidate feasibility) Every candidate regime has positive stationary probability, uniformly over Γ^{μ} and Γ^{σ} (automatic with two regimes per dimension)

Remark B2: adjacent regimes distinguishable *at* the threshold (Chan 1993, Condition 4); B3: not observationally equivalent away from it. The cell clause and B9 exist because two partitions are identified at once

Back

Theorems 1 and 2(i): Consistency and Rates

Theorem 1. *Under A1–A4, B1–B4 and B9, if the data are generated by the model at θ_0, the feasible estimator satisfies*

$$\hat{\theta}_T \to_{a.s.} \theta_0 \quad \text{as} \quad T \to \infty, \qquad \text{so} \quad (\hat{\gamma}^{\mu}, \hat{\gamma}^{\sigma}) \to_{a.s.} \left(\gamma^{\mu,0}, \gamma^{\sigma,0}\right)$$

Theorem 2 (i). *Under A1–A4 and B1–B9,*

Super-consistency of the thresholds, for every i, ℓ:

$$T\left(\hat{\gamma}_i^{\mu} - \gamma_i^{\mu,0}\right) = O_p(1), \quad T\left(\hat{\gamma}_{\ell}^{\sigma} - \gamma_{\ell}^{\sigma,0}\right) = O_p(1)$$

$\sqrt{T}$-consistency of the regime parameters:

$$\sqrt{T}\left(\hat{B}_i - B_i^0\right) = O_p(1), \quad \sqrt{T}\,\mathrm{vech}\left(\hat{\Sigma}_{\ell} - \Sigma_{\ell}^0\right) = O_p(1)$$

Proof Sketches Back

Proof Sketch for Theorems 1 and 2(i)

Consistency (extremum estimator, on the feasible profile criterion)

1. ULLN over Θ: geometric ergodicity and fourth moments (Theorem A1) give mixing and an integrable envelope
2. Unique population maximizer: a wrong threshold misassigns a set of positive probability where the regime laws differ (B2–B3): positive KL loss in each dimension
3. Argmax theorem (Newey–McFadden 1994); the one-step feasible profile is uniformly valid (Lemma T5-F; where B9 enters)

Rates (localization over the box of threshold pairs)

1. Local counting in both bands: $O_p(1)$ observations change cell when $(\gamma^{\mu}, \gamma^{\sigma})$ moves by $O(1/T)$
2. Each switched observation pays a positive expected KL loss (mean clause of B2 in the mean band, variance clause in the other)
3. Remainder: re-estimated regime parameters move by $O_p(1/T)$ (Lemma 7) $\Rightarrow$ $o_p(1)$ after scaling by T
4. $\sqrt{T}$ for the regime parameters: $O_p(1)$ misclassified observations add $O_p(T^{-1/2})$ to the score; standard expansion within each regime

Back

Assumptions for the Asymptotic Distribution

Regularity local to the thresholds; $\Lambda_t := 1 + \|x_t\|^2 + \|y_t\|^2$ is the moment envelope; B8 has no single-threshold counterpart

B5 (Local ρ-mixing, both thresholds) The process stacking $\left(\Lambda_t, q^{\mu}_{t-d_\mu}\right)$ and $\left(\Lambda_t, q^{\sigma}_{t-d_\sigma}\right)$, localized to Δ-bands around $\gamma^{\mu,0}$, $\gamma^{\sigma,0}$, is ρ-mixing with summable coefficients

B6 (Conditional moments near both thresholds) $\mathbb{E}\left[\Lambda_t^2 \mid q^{\mu}_{t-d_\mu} = v\right] \le M$ and $\mathbb{E}\left[\Lambda_t^2 \mid q^{\sigma}_{t-d_\sigma} = v\right] \le M$ for v near each threshold

B7 (Weak continuity, mean threshold only) Conditional law of $\left(x_t^{(-)}, \text{variance-regime indicators}\right)$ given $q^{\mu}_{t-d_\mu} = v$ weakly continuous at $\gamma^{\mu,0}$; the weights $p_\ell = P\left(q^{\sigma}_{t-d_\sigma} \in \mathcal{R}_\ell^{\sigma,0} \mid q^{\mu}_{t-d_\mu} = \gamma^{\mu,0}\right)$ well-defined

- ! Not needed at $\gamma^{\sigma,0}$: variance jumps depend on ε_t alone ($\perp \mathcal{F}_{t-1}$)

B8 (Bounded joint density) $\left(q^{\mu}_{t-d_\mu}, q^{\sigma}_{t-d_\sigma}\right)$ admits a joint density bounded near $\left(\gamma^{\mu,0}, \gamma^{\sigma,0}\right)$

- Price of localizing two thresholds at once; automatic for distinct coordinates of y_t; shared state with equal thresholds = joint model

Back

Theorem 2(ii)–(iii): Limiting Distributions

Jump variables (reassign one regime, keep the other) — variance jumps free of x_t:

$$\xi_t^{\mu,+} \big| (x_t, \ell(t)) \sim N\left(-\tfrac{1}{2}\kappa_t,\ \kappa_t\right), \qquad \kappa_t = \delta_t^{\mu\prime} \Sigma_{\ell(t)}^{0,-1} \delta_t^{\mu}$$

$$\xi_t^{\sigma,+} \xrightarrow{d} \tfrac{1}{2} \sum_{j=1}^{n} \left[\ln \lambda_j + (1-\lambda_j) Z_j^2\right], \qquad Z_j \overset{iid}{\sim} N(0,1), \quad \lambda_j = \text{eig}\left(\Sigma_1^{0,-1} \Sigma_2^0\right)$$

Theorem 2 (ii)–(iii). *Under A1–A4 and B1–B9 (two regimes shown):*

(ii) $\left(T\left(\hat{\gamma}^{\mu} - \gamma^{\mu,0}\right), T\left(\hat{\gamma}^{\sigma} - \gamma^{\sigma,0}\right)\right) \Rightarrow (M^{\mu}, M^{\sigma})$: *independent compound-Poisson argmax limits; mean-jump law a mixture over variance regimes (weights p_ℓ)*

(iii) *Independent of the thresholds, as if they were known:*

$$\sqrt{T}\,\text{vec}\left(\hat{\Phi}_i^{GLS} - \Phi_i^0\right) \Rightarrow N\left(0,\ (M_i^{GLS})^{-1}\right)$$

$$\sqrt{T}\,\text{vech}\left(\hat{\Sigma}_\ell - \Sigma_\ell^0\right) \Rightarrow N\left(0,\ \tfrac{2}{\pi_\ell^\sigma} D_n^+ \left(\Sigma_\ell^0 \otimes \Sigma_\ell^0\right) D_n^{+\prime}\right)$$

with all Gaussian blocks mutually independent; one-step feasible GLS is first-order equivalent to infeasible GLS

$M_i^{GLS} = \mathbb{E}_\pi\left[x_t x_t' \otimes \Sigma_{\ell(t)}^{0,-1} \mathbb{I}\{q_{t-d_\mu}^{\mu} \in \mathcal{R}_i^{\mu,0}\}\right]$, $\pi_\ell^\sigma = P_\pi\left(q_{t-d_\sigma}^{\sigma} \in \mathcal{R}_\ell^{\sigma,0}\right)$. **GLS efficiency:** $(M_i^{GLS})^{-1} \preceq$ OLS sandwich variance.

Proof Sketch Back

Proof Sketch for Theorem 2(ii)–(iii)

- **Reduction** (Lemma 7): the feasible local process equals the true-parameter one up to $o_p(1)$ on compacts
- **Two local processes**: both thresholds rescaled at $1/T$; mean-side jumps $\xi^{\mu,\pm}$, variance-side jumps $\xi^{\sigma,\pm}$, each from $O_p(1)$ boundary obs.
- **Disjointness (B8)**: $P(\text{obs. within } O(1/T) \text{ of } \textit{both} \text{ thresholds}) = O(1/T) \Rightarrow$ the two processes are driven by disjoint boundary sets $\Rightarrow$ the limiting Poisson random measure factorizes: asymptotic independence
- **Marks**: mean side as in the joint model but a mixture over variance regimes (weights p_ℓ, B7); variance side i.i.d. functions of ε_t alone, explicit law, no continuity condition
- **Argmax**: all jump means $= -\mathrm{KL} < 0$, both walks drift to $-\infty$, a.s. unique maximizing intervals; continuous mapping
- **Regime parameters**: misclassification $O_p(1)$; score expansion: one-step FGLS = infeasible GLS $+ o_p(T^{-1/2}) \Rightarrow$ Gaussian limits at the GLS bound

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